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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
import numpy as np

iris = load_iris()

X = iris.data
y = iris.target

print("IRIS CLASSIFICATION USING NAIVE BAYES")
print("-----")

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("\nActual Values:")
print(y_test)

print("\nPredicted Values:")
print(y_pred)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,

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        target_names=iris.target_names
    ))

new_flower = np.array([[5.8, 2.7, 4.1, 1.0]])

prediction = model.predict(new_flower)

probability = model.predict_proba(new_flower)

print("\nNew Flower Prediction:")
print(iris.target_names[prediction[0]])

print("\nPrediction Probabilities:")
print(probability)

IRIS CLASSIFICATION USING NAIVE BAYES
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..
Actual Values:
[0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]

Predicted Values:
[0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 2 1 0 2 0]

Accuracy:
0.9666666666666667

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]

Classification Report:
Classification Report:
      precision    recall  f1-score   support

setosa         1.00      1.00      1.00        10
versicolor     1.00      0.90      0.95        10
virginica       0.91      1.00      0.95        10

accuracy              0.97              30
macro avg            0.97      0.97      0.97        30
weighted avg         0.97      0.97      0.97        30

New Flower Prediction:
versicolor

Prediction Probabilities:
[[5.12378164e-70  9.99990961e-01  9.03882552e-06]]

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